Supplementary Note: ATAC-seq to Gene Programs and Gene Sets in dig-gene-set-extractors

Overview and goals

The `dig-gene-set-extractors` repository provides a suite of converters that map diverse omics data types to: (i) *gene programs* (scores or weights over genes) and (ii) compact *gene sets* (100–500 genes) suitable for enrichment and mechanism-level interpretation.

This note focuses on ATAC-seq converters. ATAC-seq measures chromatin accessibility at genomic loci, so any gene-level representation is a model-based projection from locus space to gene space. In practice, users rarely want "genes with the most open chromatin" in an absolute sense; they typically want gene sets that reflect the *biological question that motivated generating ATAC-seq*, such as:

- cell type or cell state identity programs (scATAC clusters),
- condition or phenotype-associated regulatory changes (opening vs closing),
- enhancer-driven (distal) regulation rather than broadly open promoters,
- upstream regulator (TF) programs,
- and (optionally) reference-calibrated tissue specificity or atlas matching.

Accordingly, the ATAC-seq pipeline is designed as a *program factory*: for a given dataset (and optionally for each group/cluster), it can emit multiple named program instances. Each program instance is then converted to one or more gene sets and exported as `.gmt`.

Default behavior and extensibility

The implementation supports two usage modes:

- **Sensible defaults**: run a small number of high-value, low-dependency program methods that work broadly across datasets.
- **All available methods**: run all program methods that are requested, skipping or failing on methods whose auxiliary resources are missing according to user preference.

This note is organized to match that workflow: we define a unified scoring framework and then describe a catalog of program methods, grouped by motivation and data requirements, including the auxiliary resources needed for each method.

1 Mental model of ATAC-seq for gene program extraction

ATAC-seq measures chromatin accessibility at genomic loci. The primary measured objects are regions (peaks) with coordinates and a locus-level signal. Many downstream biological questions, however, are gene-centric (pathways, processes, mechanisms), so we require an explicit model that projects locus-level evidence to genes.

Promoters vs enhancers. Promoter accessibility is often broadly shared across cell types and can be dominated by global effects (coverage, cell cycle, stress, housekeeping). Distal enhancer accessibility is typically more informative for cell type and cell state identity. For this reason, the default program preset emphasizes distal programs and uses reference calibration when available.

Baseline vs contrast. A single-sample peak list (baseline accessibility) often yields generic programs unless we (i) emphasize distal enhancers and/or (ii) calibrate against reference accessibility. Stronger specificity is usually obtained from within-study contrasts (for example, cluster-vs-rest in scATAC or condition A vs B across donors). This note distinguishes: (i) study-design contrasts that define the peak statistic x_p and (ii) calibration methods (`contrast_methods`) that adjust peak/gene signals relative to external baselines.

What a contributor typically has. ATAC-seq contributors commonly provide one of:

- **Bulk peak intervals** (BED/narrowPeak) with an optional signal column (converter: `atac_bulk`).
- **scATAC peak-by-cell matrix** (counts) plus group labels (converter: `atac_sc_10x`).
- **Bulk peak-by-sample matrix** plus sample metadata defining a contrast (converter: `atac_bulk_matrix`).

Pipeline summary. Across these inputs, the core pipeline is: (1) define a peak statistic x_p, (2) transform to a nonnegative peak weight $\alpha_p = \phi(x_p)$, (3) optionally calibrate peaks/genes against references (contrast-method axis), (4) project to genes via a linkage model L_{pg} to obtain gene scores s_g, (5) apply a program-family view (linked/promoter/distal/etc.), and (6) convert gene scores to one or more gene sets.

2 How to map CLI arguments to the model in this note

This supplementary note provides the biological rationale and mathematical definitions for the components exposed as command line arguments in `omics2geneset`. Practical usage examples and full flag listings are in `docs/omics2geneset.md`. The two documents are designed to be read in either order.

CLI flags (guide)	Model object (this note)	Defined in this note
`--peaks`, `--peak_matrix_tsv`, `--matrix_dir`	Peak coordinates and peak statistic x_p	Sections 1, 4.1
`--gtf`, `--organism`, `--genome_build`	Gene coordinates (TSS/promoter/locus)	Section 6
`--contrast` and related condition flags	Study-design contrast defining x_p	Section 7

`--peak_weight_transform`	Transform $\alpha_p = \phi(x_p)$	Section 4.1
`--contrast_methods`	External calibration (peak-level or gene-level)	Section 8, Eqs. 6, 7, 8
`--ref_ubiquity_resource_id`, `--atlas_resource_id`, `--atlas_metric`, `--atlas_min_raw_quantile`, `--atlas_use_log1p`	Reference resources and stabilization choices	Section 14, Subsections M3.1 and M3.2
`--link_method`	Linkage model L_{pg}	Section 6
`--program_preset`, `--program_methods`	Program-family view Ψ_m	Sections 9, 10
`--select`, `--top_k`, `--normalize`	Gene set extraction operator and normalization	Section 11, Eq. 10
`--emit_gmt`, `--gmt_topk_list`, `--gmt_min_genes`, `--gmt_max_genes`	GMT export of selected gene sets	Section 12

3 Notation

- Peaks (regions) indexed by $p \in \{1, \ldots, P\}$ with coordinates (c_p, s_p, e_p) and midpoint $m_p = (s_p + e_p)/2$.
- Genes indexed by $g \in \{1, \ldots, G\}$ with chromosome c_g, strand $\sigma_g \in \{+, -\}$, TSS position t_g, promoter interval $\mathcal{P}_g$, and gene locus interval $\mathcal{G}_g$.
- In bulk ATAC, samples indexed by $i \in \{1, \ldots, n\}$.
- In scATAC, cells indexed by $c \in \{1, \ldots, C\}$, with peak-by-cell matrix $X_{pc} \geq 0$ (counts).
- A group/cluster label function $\gamma(c) \in \{1, \ldots, K\}$, where $\mathcal{C}_k = \{c : \gamma(c) = k\}$.
- A peak-level statistic x_p (absolute accessibility, differential statistic, association statistic, or other).
- A peak weight $\alpha_p \geq 0$ derived from x_p via a transform ϕ (Sections 4.1 and 10.4).
- A peak-to-gene linkage weight $L_{pg} \geq 0$ (Section 6).
- A gene score s_g (real-valued or nonnegative) and optionally a normalized distribution w_g.

4 Unified scoring framework

4.1 Peak weights

Define a transform ϕ that maps a peak statistic x_p to a peak weight α_p:

$$\alpha_p = \phi(x_p).$$

Common transforms:

$$\begin{aligned}\phi_{\text{signed}}(x) &= x,\\ \phi_{\text{abs}}(x) &= |x|,\\ \phi_{\text{pos}}(x) &= \max(x, 0),\\ \phi_{\text{neg}}(x) &= \max(-x, 0).\end{aligned}$$

For directional programs (opening vs closing), apply ϕ_{pos} and ϕ_{neg} separately to obtain two non-negative peak-weight vectors. CLI: `--peak_weight_transform` selects ϕ; study-design contrasts provide x_p (Section 7).

4.2 Peak-to-gene projection

Given peak weights α_p and link weights L_{pg}, define a raw gene score:

$$s_g \;=\; \sum_{p=1}^{P} \alpha_p \, L_{pg}. \tag{1}$$

Equation (1) is the core projection used throughout. Biological meaning is determined by: (i) how x_p is defined (absolute vs contrast vs association), and (ii) how L_{pg} is defined (promoter vs distal vs learned wiring).

4.3 From scores to distributions (optional)

If $s_g \geq 0$ and not all zero, a global L1-normalized distribution is:

$$w_g \;=\; \frac{s_g}{\sum_{h=1}^{G} s_h}, \qquad \sum_{g=1}^{G} w_g = 1. \tag{2}$$

This is a *relative allocation* across genes, not a per-gene probability of activity. When the goal is compact programs (100–500 genes), global normalization can make weights appear numerically small and visually uniform if many genes receive nonzero score. Therefore, gene set extraction (Section 11) is an explicit step, and within-program normalization is preferred for interpretability.

5 Conceptual extraction target and configurable axes

ATAC-seq measures accessibility at regulatory regions, but downstream interpretation is usually gene-centric (pathways, perturbation signatures, cross-modality comparison). The extraction target is therefore a *gene-level regulatory program*:

1. represent locus-level evidence in a way aligned to the biological question,
2. project region evidence to genes through an explicit linking model,
3. derive one or more program views, and
4. export compact gene sets for downstream analyses.

In implementation terms, the gene sets are generated over three explicit, orthogonal axes:

- **Link axis** (ℓ): how peaks are linked to genes, via $L_{pg}^{(\ell)}$. CLI: `--link_method`. Output tag: `__link_method=...`
- **Contrast-method axis** (r): how raw accessibility is calibrated against baseline/reference context. CLI: `--contrast_methods`. Output tag: `__contrast_method=...`
- **Program-family axis** (m): what biological program view is emitted from linked scores (linked/promoter/distal/enhancer-bias/TFIDF variants). CLI: `--program_methods` or `--program_preset`. Output tag: `__program=...`

Program families are not merely aliases for link choices: they can apply additional operations (for example promoter/distal peak masking or distal-vs-promoter combination) before/after projection. In the `all` preset they are evaluated across selected link models; in recommended presets, execution is intentionally constrained to high-value program/link pairs.

Given these axes, the shared projection is:

$$s_g^{(r,\ell)} = \sum_p \alpha_p^{(r)} L_{pg}^{(\ell)},$$

and each program family produces a gene score view

$$u_g^{(m,r,\ell)} = \Psi_m\Big(s^{(r,\ell)}, \alpha^{(r)}, L^{(\ell)}\Big),$$

followed by the set selection operators in Section 11.

6 Peak-to-gene linkage models

The linkage model encodes biological assumptions about cis-regulatory wiring. All linkage models can be combined with any definition of x_p and any gene set extraction operator.

6.1 Link A: promoter overlap

Biology. Promoter accessibility is directly connected to transcriptional competence and is minimally ambiguous to assign.

Definition.

$$L_{pg} = \mathbf{1}\{c_p = c_g \wedge [s_p, e_p) \cap \mathcal{P}_g \neq \emptyset\}.$$

Heuristics. Promoter window (u, d) around the TSS (e.g., $u = 2000$ bp upstream, $d = 500$ bp downstream). Record whether transcript union or a canonical TSS is used.

6.2 Link B: nearest TSS assignment

Biology. A simple distal heuristic: regulatory elements often regulate a nearby gene.

Definition. With distance $d(p, g) = |m_p - t_g|$ and max distance D:

$$g^*(p) = \arg\min_{g:d(p,g)\leq D} d(p, g), \qquad L_{pg} = \mathbf{1}\{g = g^*(p)\}.$$

Heuristics. Use per-chromosome sorted TSS arrays for $O(\log G)$ nearest lookup. Tie-break deterministically by gene ID.

6.3 Link C: distance-decay soft assignment

Biology. Contact probability and average regulatory influence decrease with genomic distance.

Definition. For decay length λ and max distance D:

$$A_{pg} = \begin{cases} \exp(-d(p,g)/\lambda), & d(p,g) \le D \\ 0, & \text{otherwise.} \end{cases}$$

Either use $L_{pg} = A_{pg}$ (unnormalized), or normalize per peak:

$$L_{pg} = \frac{A_{pg}}{\sum_h A_{ph} + \epsilon}.$$

Heuristics. Defaults such as $D = 500$ kb, $\lambda = 50$ kb; optionally cap to the top $K_{\max}$ genes per peak by A_{pg}.

6.4 Link D: gene locus activity (gene body plus promoter)

Biology. Aggregating accessibility across promoter and gene body yields a gene-activity proxy useful in sparse scATAC.

Definition. Define a gene locus interval $\mathcal{G}_g$ and compute:

$$s_g = \sum_{p:[s_p,e_p)\cap\mathcal{G}_g\neq\emptyset} \alpha_p.$$

Heuristics. This can induce gene-length bias. Mitigate by standardized windows, length normalization, or by using this as a complementary program rather than the sole output.

6.5 Link E: co-accessibility linkage

Biology. Peaks that covary across cells can reflect enhancer-promoter coupling.

Definition. Let C_{pq} be a peak-peak co-accessibility score. For each gene g with promoter peak(s) $q \in \mathcal{Q}(g)$:

$$L_{pg} = \max\left(\max_{q\in\mathcal{Q}(g)} C_{pq},\, 0\right) \cdot \mathbf{1}\{d(p,g) \le D\}.$$

Heuristics. Requires many cells. Restrict candidate pairs to windows (e.g., 250–500 kb), threshold weak edges, and record co-accessibility method.

6.6 Link F: external enhancer-gene priors

Biology. External enhancer-gene maps integrate multi-assay evidence and can improve specificity.

Definition. Overlap peaks to external elements e with a link matrix M_{eg}:

$$L_{pg} = \sum_e \mathbf{1}\{p \cap e \neq \emptyset\} M_{eg}.$$

Heuristics. Choose a default map per genome build; record version and provenance. Handle biosample mismatch explicitly (closest match vs union vs average).

6.7 Link G: activity-by-contact style linkage (optional)

Biology. Enhancer effect is proportional to enhancer activity and contact with a promoter.

Definition. With an activity estimate A_p (from ATAC or multi-assay) and a contact kernel K_{pg}:

$$S_{pg} = A_p K_{pg}, \qquad L_{pg} = \frac{S_{pg}}{\sum_h S_{ph} + \epsilon}.$$

Heuristics. If no contact map is available, approximate K_{pg} by a distance kernel (exponential or power-law), and record the approximation.

7 Study-design contrasts (defining the peak statistic x_p)

Before calibration by `--contrast_methods`, users define a study-design contrast that produces peak statistics x_p:

- baseline accessibility (no within-study contrast),
- group-vs-rest (scATAC),
- condition-within-group (scATAC),
- condition A vs B across samples (bulk matrix),
- phenotype association (optional regression-based workflows).

These design-level contrasts define the signed or unsigned peak statistics that are then transformed to peak weights and passed through the rest of the pipeline. In output naming this study-design stage appears as `__contrast=...`.

Study-design contrast defines x_p first. The calibration contrast-method axis (`--contrast_methods`) is applied after x_p is defined, and adjusts peak or gene signals relative to external references.

8 Contrast-method axis

The contrast-method axis (r) defines how peak or gene signals are calibrated before extracting gene sets. Conceptually, this axis answers: "contrast against what baseline, and at what stage (peak-level or gene-level)?"

8.1 Contrast method options

- `none`: no external calibration; use raw within-dataset peak statistics.
- `ref_ubiquity_penalty`: reweight peaks by reference ubiquity (peak-level calibration) before projection.
- `atlas_residual`: calibrate projected gene scores against reference atlas statistics (gene-level calibration).

9 Program-family axis

The program-family axis (m) defines which biological view is emitted from a given linked score space. This axis answers: "what kind of gene program do we want to report from the same underlying contrast and linkage choice?"

9.1 Program-family options

- `linked_activity`: direct projection of linked scores.
- `promoter_activity`: promoter-only view.
- `distal_activity`: distal-only view.
- `enhancer_bias`: distal-vs-promoter residualized view.
- `tfidf_distal` (scATAC): specificity-weighted distal view.

Program families are independent of link and contrast-method choices, and are evaluated for each active (r, ℓ) combination. The "method families" in Section 10 are practical bundles of axis choices for common biological questions; they are not additional axes.

10 ATAC program methods catalog

This section defines a catalog of program methods. Each method corresponds to a biological motivation and specifies: (i) how peak weights α_p are constructed, (ii) which peaks are included (promoter vs distal), (iii) which linkage model L_{pg} is used, and (iv) what program instances are emitted (one per sample, per group, per direction, per module, etc.).

Implementation axis decomposition. As defined in Sections 5, 8, and 9, CLI outputs are generated over three independent axes:

- `link_method`: region-to-gene linkage model (promoter overlap, nearest TSS, distance decay, external).
- `contrast_methods`: calibration/contrast choice (`none`, `ref_ubiquity_penalty`, `atlas_residual`).
- `program_methods`: program-family view (`linked_activity`, `promoter_activity`, `distal_activity`, `enhancer_bias`, and `tfidf_distal` where applicable).

Each active contrast method is evaluated for each selected linkage model, and each resulting linked score space is then emitted through requested program families. The method families below are integrative templates built from these three axes.

10.1 Summary table

Table 2 summarizes the most important methods, their typical use cases, and resource requirements.

Table 2: ATAC program methods (non-exhaustive). Defaults are intended to be high-value and low-dependency.

Method ID	What it captures	Requires	Default?
promoter_activity	Broad promoter accessibility (QC / global activity)	Peaks + GTF	QC only
distal_activity	Enhancer-driven accessibility (more tissue-specific)	Peaks + GTF	Yes
enhancer_bias	Distal vs promoter residual (enhancer emphasis)	Peaks + GTF	Experimental
group_vs_rest_open	Cluster-specific opening (scATAC)	scATAC + groups	Yes
group_vs_rest_close	Cluster-specific closing (scATAC)	scATAC + groups	Optional
tfidf_distal	TF-IDF specificity-weighted distal cluster program	scATAC matrix	Experimental
ref_ubiquity_penalty	Penalize elements open in many tissues/cells	Reference cCRE/atlas	Optional
atlas_residual	Sample-specific vs atlas baseline	Reference atlas	Optional
motif_regulons	TF driver programs via motif hits	Motifs + genome/hits	Optional
coaccess_modules	Peak communities -¿ gene modules	Many cells	Optional
topic_programs	Latent topics (NMF/LDA) -¿ gene programs	Many cells	Optional
phenotype_open/close	Condition/phenotype differential programs	Multi-sample labels	Optional
variant_anchored	Variant-overlap anchored gene programs	Variant set	Optional

10.2 Method family 1: absolute activity programs

These methods operate on an absolute accessibility statistic per peak and are most appropriate for: (i) multi-sample studies (where differential methods can be used) or (ii) quality control summaries. For single-sample bulk ATAC, absolute promoter activity tends to yield broadly open chromatin genes; distal and specificity-weighted programs are preferred.

10.2.1 M1.1: promoter_activity

Peak weights. Choose an absolute peak statistic x_p (e.g., normalized coverage, peak score, or group summary). Set $\alpha_p = \phi_{\text{pos}}(x_p)$ to ensure nonnegativity.

Peak filter. Keep only peaks overlapping any promoter interval $\mathcal{P}_g$.

Link. Use Link A (promoter overlap).

Gene score. Compute s_g by Eq. 1.

Gene sets. Select top-K genes (Section 11).

10.2.2 M1.2: distal_activity

Motivation. Distal regulatory elements are often more tissue- and state-specific than promoters.

Peak weights. Same as promoter_activity.

Peak filter. Keep only distal peaks:

$$\mathcal{D} = \{p : \forall g,\ [s_p, e_p) \cap \mathcal{P}_g = \emptyset\},$$

or equivalently peaks with distance to nearest TSS greater than a chosen promoter radius.

Link. Use Link C (distance-decay) or Link B (nearest TSS) for distal assignment.

Gene score and sets. As above.

10.2.3 M1.3: enhancer_bias (distal vs promoter residual)

Motivation. A gene may be broadly expressed (open promoter) but not enhancer-driven in the sample. This method prioritizes genes whose distal accessibility is high relative to promoter accessibility.

Definition. Compute two nonnegative scores:

$$s_g^{\text{prom}} = \sum_{p \in \text{promoters}} \alpha_p L_{pg}, \qquad s_g^{\text{dist}} = \sum_{p \in \text{distal}} \alpha_p L_{pg}.$$

Define an enhancer-bias score:

$$b_g = \log\left(\frac{s_g^{\text{dist}} + \epsilon}{s_g^{\text{prom}} + \epsilon}\right). \tag{3}$$

Alternatively, normalize each component across genes before differencing:

$$b_g = \log(s_g^{\text{dist}} + \epsilon) - \log(s_g^{\text{prom}} + \epsilon), \quad \text{or} \quad b_g = z(s_g^{\text{dist}}) - z(s_g^{\text{prom}}).$$

Gene sets. Select top-K genes by b_g.

10.3 Method family 2: contrasts and specificity (within-study)

These methods are designed for the most common scientific question: *what differs across groups, conditions, or cell states?*

10.4 scATAC peak summaries

For scATAC, define a group-level peak summary for group $\mathcal{C}_k$:

$$\text{sum counts: } \bar{x}_p^{(k)} = \sum_{c \in \mathcal{C}_k} X_{pc},$$

$$\text{mean counts: } \bar{x}_p^{(k)} = \frac{1}{|\mathcal{C}_k|} \sum_{c \in \mathcal{C}_k} X_{pc},$$

$$\text{fraction nonzero: } \bar{x}_p^{(k)} = \frac{1}{|\mathcal{C}_k|} \sum_{c \in \mathcal{C}_k} \mathbf{1}\{X_{pc} > 0\}.$$

10.4.1 M2.1: group_vs_rest_open (scATAC default)

Motivation. Cell identity and state are most naturally described by accessibility that is *specific* to a group vs background.

Peak statistic. For group k, let $\bar{x}_p^{(k)}$ be the group summary and $\bar{x}_p^{(\neg k)}$ be the background (all other cells) summary. Define:

$$x_p^{(k)} = \log_2 \left(\frac{\bar{x}_p^{(k)} + a}{\bar{x}_p^{(\neg k)} + a} \right), \tag{4}$$

with pseudocount $a > 0$ (e.g., $a = 10^{-3}$ for fractions, $a = 1$ for counts).

Peak weights. $\alpha_p^{(k)} = \max(x_p^{(k)}, 0)$ (opening program).

Link. Use Link A for promoter-focused cluster markers and Link C for distal-aware markers; both may be emitted.

Gene sets. Select top-K genes.

10.4.2 M2.2: group_vs_rest_close (optional)

Same as M2.1 but with $\alpha_p^{(k)} = \max(-x_p^{(k)}, 0)$ to capture peaks that are less accessible in the group than background.

10.4.3 M2.3: condition_within_group_open / condition_within_group_close

Motivation. When cells have a semantic handle (perturbation, phenotype, time, genotype), the most connectable signal is often the within-cell-type contrast, not global accessibility.

Setup. For group k, split cells by condition labels A and B and compute per-peak summaries $\bar{x}_p^{(k,A)}$ and $\bar{x}_p^{(k,B)}$ (sum/mean/fraction nonzero).

Peak statistic.

$$x_p^{(k,A:B)} = \log_2 \left(\frac{\bar{x}_p^{(k,A)} + a}{\bar{x}_p^{(k,B)} + a} \right),$$

or $x_p^{(k,A:B)} = \bar{x}_p^{(k,A)} - \bar{x}_p^{(k,B)}$ when using a difference metric.

Direction-aware programs.

$$\alpha_{p,\mathrm{OPEN}}^{(k)} = \max(x_p^{(k,AB)}, 0), \qquad \alpha_{p,\mathrm{CLOSE}}^{(k)} = \max(-x_p^{(k,AB)}, 0).$$

Project each direction to genes via Eq. 1 and extract separate OPEN and CLOSE gene sets.

Practical note. Enforce a minimum number of cells per condition within each group to avoid unstable contrasts.

10.4.4 M2.4: tfidf_distal (scATAC optional / experimental)

Motivation. Peaks that are accessible in many cells are less informative for defining cell identity. A TF-IDF style reweighting emphasizes more specific peaks. This family is not part of the connectable/default preset; enable it with `--program_preset experimental`, `--program_preset all`, or explicit `--program_methods tfidf_distal`.

Definition. Let $N = C$ be the number of cells in the dataset and define the document frequency:

$$df_p = \sum_{c=1}^{C} \mathbf{1}\{X_{pc} > 0\}.$$

For group k, define a term frequency:

$$tf_p^{(k)} = \sum_{c \in \mathcal{C}_k} X_{pc} \quad \text{or} \quad tf_p^{(k)} = \sum_{c \in \mathcal{C}_k} \mathbf{1}\{X_{pc} > 0\}.$$

Define:

$$idf_p = \log\left(\frac{N+1}{df_p+1}\right) + 1, \qquad \alpha_p^{(k)} = \log(1 + tf_p^{(k)}) \cdot idf_p. \tag{5}$$

Gene projection and sets. Apply Eq. 1 and top-K selection. As with M2.1, both promoter-only and distal-only variants may be emitted.

Status. In current CLI defaults (`program_preset=connectable/default`), this family is not emitted by default; enable it via `program_preset=experimental` or explicit `--program_methods tfidf_distal`.

10.4.5 M2.5: phenotype_open / phenotype_close (bulk or sc, optional)

Motivation. Many ATAC studies compare donors/conditions (case vs control, treatment, time, severity).

Peak statistic. Suppose we have per-sample peak measurements $x_{p,i}$ and a phenotype y_i. Fit a regression model per peak:

$$x_{p,i} = \beta_{p,0} + \beta_{p,1} y_i + \mathbf{z}_i^\top \boldsymbol{\gamma}_p + \varepsilon_{p,i},$$

where $\mathbf{z}_i$ are covariates. Let t_p or z_p be a signed test statistic for $\beta_{p,1}$.

Peak weights. Opening: $\alpha_p^+ = \max(z_p, 0)$; closing: $\alpha_p^- = \max(-z_p, 0)$.

Gene projection. Use any linkage model (promoter or distal-aware).

Gene sets. Emit top-K genes for opening and closing.

10.5 Method family 3: reference-calibrated specificity (optional resources)

Absolute accessibility and within-study contrasts do not always provide tissue specificity for single-sample bulk ATAC. Reference-based calibration addresses this.

10.5.1 M3.1: ref_ubiquity_penalty (reference IDF)

Motivation. Penalize peaks that are open in many tissues/cell types; emphasize tissue- or state-specific regulatory DNA.

Inputs. A reference registry of regulatory elements (e.g., a cCRE catalog) with an "accessibility ubiquity" statistic across reference biosamples.

Definition. Let $df^{\text{ref}}(e)$ be the number of reference biosamples in which element e is accessible, and N_{ref} the number of reference biosamples. Define:

$$idf^{\text{ref}}(e) = \log\left(\frac{N_{\text{ref}} + 1}{df^{\text{ref}}(e) + 1}\right) + 1.$$

If a peak p overlaps element e, modify its weight:

$$\alpha_p^{\text{adj}} = \alpha_p \cdot idf^{\text{ref}}(e), \tag{6}$$

or sum over multiple overlapping elements. Then project to genes by Eq. 1.

Heuristics. If no overlap is found, use $idf^{\text{ref}} = 1$. Record the reference catalog version and the exact definition of df^{ref}.

10.5.2 M3.2: atlas_residual (gene-level reference residualization)

Motivation. Rather than reweight peaks, calibrate at the gene-score level relative to an atlas of reference tissues/cell types.

Inputs. A reference panel of gene scores $s_g^{(t)}$ for reference context t (precomputed using the same projection model).

Definitions. Let $m_g = \text{median}_t(s_g^{(t)})$ and $d_g = \text{MAD}_t(s_g^{(t)})$ (median absolute deviation). Define either:

$$r_g = \log\left(\frac{s_g + \epsilon}{m_g + \epsilon}\right), \tag{7}$$

$$z_g = \frac{s_g - m_g}{d_g + \epsilon}. \tag{8}$$

Heuristics and safeguards. Atlas baselines should be computed with the same score definition (program family, link method, and related scoring choices) as the run being calibrated; otherwise the residual is not valid and should be skipped or treated as an error. To reduce near-zero denominator artifacts, apply a raw-score floor (for example, keep genes above a raw-score quantile before ranking residuals) and/or compute residuals on log1p scores. In the CLI these controls are exposed by `--atlas_min_raw_quantile` and `--atlas_use_log1p`; `--atlas_metric` chooses `logratio` versus `zscore`.

Gene sets. Select top-K genes by r_g or z_g to obtain a reference-calibrated specificity program.

10.6 Method family 4: regulator-centric programs (optional resources)

Implementation status. Described as a forward-looking extension; not enabled by default and may not be implemented in the current release.

10.6.1 M4.1: motif_regulons

Motivation. A common ATAC question is "which TFs are driving the regulatory program?" Motif-linked programs provide a gene-level readout of TF activity hypotheses.

Inputs. A motif database and motif occurrences in peaks (computed or precomputed).

Definition. For motif m, let $\mathcal{P}(m)$ be the peaks containing that motif (optionally weighted by match score). Define a motif-specific gene score:

$$s_g^{(m)} = \sum_{p \in \mathcal{P}(m)} \alpha_p \, L_{pg}. \tag{9}$$

Program selection. To avoid producing hundreds of sets, restrict to motifs with high total mass $\sum_g s_g^{(m)}$ or motifs enriched among top peaks. Emit top-K genes per selected motif.

10.7 Method family 5: module discovery (optional, heavier compute)

Implementation status. Described as a forward-looking extension; not enabled by default and may not be implemented in the current release. These methods produce *multiple* programs per dataset by discovering regulatory modules in peak space and projecting them to genes.

10.7.1 M5.1: coaccess_modules

Inputs. scATAC peak-by-cell matrix with sufficient cell count.

Definition. Compute a co-accessibility graph $G = (V, E)$ where nodes are peaks and edges connect peaks with $C_{pq} > \theta$ within a distance window. Cluster G into modules $\mathcal{M}_1, \ldots, \mathcal{M}_M$. For module k, define peak weights:

$$\alpha_p^{(k)} = \alpha_p \cdot \mathbf{1}\{p \in \mathcal{M}_k\}.$$

Project to genes by Eq. 1 to obtain $s_g^{(k)}$, then select top-K genes to define a gene program per module.

10.7.2 M5.2: topic_programs (NMF/LDA)

Inputs. scATAC peak-by-cell matrix (typically TF-IDF normalized).

Definition. Fit a nonnegative factorization:

$$X \approx WH,$$

where W is peak-by-topic and H is topic-by-cell. For topic k, define peak weights $\alpha_p^{(k)} = W_{pk}$ and project to genes:

$$s_g^{(k)} = \sum_p W_{pk} L_{pg}.$$

Select top-K genes per topic to obtain topic gene sets.

10.8 Method family 6: variant-anchored programs (optional)

10.8.1 M6.1: variant_anchored

Motivation. ATAC-seq is often used to interpret noncoding variants (GWAS fine-mapping, QTL) by intersecting variants with accessible chromatin.

Inputs. A set of variants with optional weights (e.g., posterior inclusion probabilities, PIPs) and genome build matching the peaks.

Definition. Let $\mathcal{V}(p)$ be variants overlapping peak p. Define:

$$\alpha_p = \sum_{v \in \mathcal{V}(p)} w_v,$$

with $w_v = 1$ if unweighted. Project to genes by Eq. 1 using a distal-aware linkage (Link C/F/G). Select top-K genes to obtain a variant-anchored gene program.

11 From gene scores to gene sets

After the three axes and program catalog determine a gene-score vector for each program instance, this section specifies how scores are converted to one or more gene sets of size 100–500. CLI: `--select`/`--top_k` define the selection operator; `--normalize` defines within-set normalization; `--emit_gmt` controls GMT export.

Practical default. The default target size is 100–500 genes (`--gmt_min_genes=100`, `--gmt_max_genes=500`). If a program instance has too few positive genes after filtering, that GMT entry is skipped by default and a warning is emitted. Users can opt in to writing smaller sets with `--emit_small_gene_sets true`.

Conceptual eligibility criterion (theory). For a program instance (m, r, ℓ), define an informative-support size

$$I^{(m,r,\ell)} = \left|\left\{g : s_g^{(m,r,\ell)} > 0\right\}\right|.$$

If $I^{(m,r,\ell)}$ is well below the target set size range, the resulting set is dominated by boundary effects and is generally less stable for downstream enrichment. The default behavior therefore treats low-support program instances as *non-emittable by default*, rather than emitting very small high-variance sets.

Grouped-contrast support (theory). For grouped scATAC contrasts, stability also depends on sample size in each stratum. Let n_k be group size and, for within-group condition contrasts, $n_{k,A}, n_{k,B}$ the per-condition counts within group k. Program instances are conceptually valid for default emission only when group-level support is adequate (sufficient n_k, and sufficient $n_{k,A}, n_{k,B}$ when applicable). Exact operational thresholds are implementation details documented in `docs/omics2geneset.md`.

11.1 Selection operators

Let $\mathbf{s} = (s_1, \ldots, s_G)$ be gene scores for a program instance. The default operator is fixed-size top-K.

S1: top-K selection (default). Sort genes by decreasing score $s_{(1)} \geq \cdots \geq s_{(G)}$ with genes $g_{(1)}, \ldots, g_{(G)}$, tie-breaking deterministically by gene ID. Define:

$$\widehat{\mathcal{G}}_K = \{g_{(1)}, \ldots, g_{(K)}\}, \qquad K \in [100, 500].$$

Within-program weights (optional) are:

$$\tilde{w}_g = \begin{cases} \dfrac{\max(s_g, 0)}{\sum_{h \in \widehat{\mathcal{G}}_K} \max(s_h, 0)}, & g \in \widehat{\mathcal{G}}_K \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

If the denominator is zero, fall back to uniform weights within $\widehat{\mathcal{G}}_K$.

S2: HPD mass set. If scores are converted to a nonnegative distribution w_g (Eq. 2), an HPD mass set for target mass $\tau \in (0, 1)$ is:

$$K^*(\tau) = \min\left\{K : \sum_{i=1}^{K} w_{(i)} \geq \tau\right\}, \qquad \widehat{\mathcal{G}}_\tau = \{g_{(1)}, \ldots, g_{(K^*(\tau))}\}.$$

To enforce 100–500 genes, clamp K to $[K_{\min}, K_{\max}]$ and use $\widehat{\mathcal{G}}_K$.

S3: multi-resolution gene sets. For a fixed score vector, optionally emit multiple nested sets (e.g., $K \in \{100, 200, 500\}$) and/or disjoint tiers (e.g., ranks 1–100, 101–200, etc.) for sensitivity analysis.

11.2 Signed programs

If a program produces signed gene scores, define nonnegative components:

$$s_g^+ = \max(s_g, 0), \qquad s_g^- = \max(-s_g, 0),$$

and apply selection operators separately to obtain two gene sets (e.g., opening vs closing).

12 GMT export

We export gene sets as `.gmt` files using a simplified one-tab, space-delimited convention:

`<gene_set_name>\t<gene1> <gene2> ... <geneN>`.

Gene tokens are gene symbols when available and non-empty; otherwise stable gene IDs. Within a set, duplicates are removed while preserving order.

Gene set names. A robust default concatenates provenance fields, for example:

`<dataset_id>__<converter>__<group>__<program_method>__<link>__<select>`.

Names are sanitized to remove whitespace and path separators.

13 Program presets and execution policy

To balance usability and flexibility, program methods are grouped into presets.

13.1 Default/connectable preset (high-value, low-dependency)

The recommended default preset (`--program_preset connectable`, with `default` as alias) emits:

- **linked_activity** with `link_method=nearest_tss`.
- **distal_activity** with `link_method=distance_decay`.
- a single contrast outcome per pair using `--contrast_methods auto_prefer_ref_ubiquity_else_none` (prefer reference ubiquity calibration, otherwise fallback to `none`).

This avoids broad, low-specificity outputs in the common case while keeping dependencies minimal.

Not recommended by default.

- promoter_activity and promoter-overlap linkage outputs for primary discovery (use mainly for QC).
- enhancer_bias and tfidf_distal as routine defaults (use in exploratory/experimental settings).
- atlas_residual without score-definition-matched baselines.

13.2 QC and experimental presets

`program_preset=qc` and `program_preset=experimental` are explicit opt-ins:

- **qc:** promoter-focused diagnostics (promoter_activity with promoter-overlap linkage).
- **experimental:** broader exploratory outputs (for example enhancer_bias and atlas_residual workflows with stabilization controls).

13.3 All-methods preset (resource-aware)

An "all" preset runs all requested methods for which required resources are available in the configured resources directory. Missing resources can be handled by: (i) skipping the method while recording that it was skipped, or (ii) failing the run (fail-fast). The chosen policy is recorded in metadata.

14 Auxiliary resources and reproducibility

Many optional methods require external resources (reference catalogs, atlases, motifs). The repository adopts best practices:

- Resources are identified by stable IDs and versioned.
- Downloads are handled by a resource manager to a single directory.
- Each resource is verified by checksum (e.g., SHA256) and recorded in output metadata.
- Outputs record genome build, annotation version, and all method parameters needed for reproducibility.

Bundled vs external. Small, permissively licensed resources may be bundled in the repository. Larger resources are hosted externally (e.g., on Zenodo) and fetched on demand. Resource entries without URLs are treated as manual resources; the fetch command can skip them (recording status) unless strict behavior is requested.

Current resource-backed hooks (implemented). Two optional methods are wired in the ATAC converters:

- `ref_ubiquity_penalty`: a *peak-weight transform*. For each peak, an IDF-like multiplier from reference overlap is applied ($\alpha'_p = \alpha_p \cdot \mathrm{idf}_p$), then all program families are recomputed from α'.
- `atlas_residual`: a *gene-level residual* computed per score definition (`program` + `link_method`) against atlas baselines. Atlas rows can be global or keyed by `score_definition`.
- Atlas residual stabilization includes (i) raw-score quantile floor filtering and (ii) optional log1p transform before `logratio`/`zscore`.

Execution policy is configurable per run (`resource_policy=skip|fail`).

15 Output artifacts and metadata

Each program instance is written to its own output directory containing:

- `geneset.full.tsv`: all genes with nonzero score (or all genes, depending on implementation),
- `geneset.tsv`: the selected program-sized gene list (100–500 genes) with scores and optional within-program weights,
- `geneset.meta.json`: machine-readable metadata describing inputs, methods, parameters, and resources,

- `genesets.gmt`: one or more gene sets derived from this program instance.

For runs producing multiple program instances (e.g., many scATAC clusters, or multiple method families), a root manifest enumerates all program directories and enables validation of the full output tree.

16 Practical heuristics and QC checks

- **Genome build consistency.** Ensure peaks, variants, and reference resources use the same build (e.g., hg38 vs hg19). Mismatches often lead to low assignment rates.
- **Chromosome naming.** Handle `chr1` vs `1` consistently; record any normalization performed.
- **Promoter dominance.** If promoter_activity yields broad housekeeping programs, prefer distal_activity, enhancer_bias, TF-IDF, or reference-calibrated methods.
- **Assignment diagnostics.** Record the fraction of peaks assigned to at least one gene and the distribution of links per peak.
- **Determinism.** Ensure sorting tie-breakers are deterministic to make runs reproducible.

17 Computational complexity (high level)

For a single program instance with P peaks and G genes, projection via Eq. 1 is $O(\#\text{links})$. In distance-based models, #links is controlled by max distance and link caps per peak. Gene set extraction is dominated by sorting gene scores: $O(G \log G)$.